MUHAMMAD QAYYUM

Associate Professor of Economics, School of Economics and Management
Quanzhou University of Information Engineering, Quanzhou, Fujian, China
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Google Scholar | ResearchGate

ACADEMIC APPOINTMENTS

Associate Professor of Economics Jan 2025 – present
School of Economics and Management, Quanzhou University of Information Engineering, Quanzhou, China

Postdoctoral Researcher in Economics Sep 2022 – Dec 2024
School of Economics & Statistics, Guangzhou University, Guangzhou, China

Postdoctoral Researcher in Economics Sep 2020 – Aug 2022
School of Economics, Hainan University, Haikou, China

EDUCATION

Ph.D., World Economy 2016 – 2020
Zhongnan University of Economics and Law, Wuhan, China
Dissertation: Machine Learning Methods, Gravity Model and Robust Determinants of Bilateral Trade
Advisor: Prof. Zhu Yanfu

M.Phil., Econometrics 2013 – 2016
Pakistan Institute of Development Economics, Islamabad, Pakistan
Thesis: Influence of Distance, Culture and Perceived Corruption on Inbound Tourism Flows in Pakistan
Advisor: Dr. Saqlain Raza

B.Sc. (Hons), Agricultural and Resource Economics 2009 – 2013
University of Agriculture, Faisalabad, Pakistan

RESEARCH INTERESTS

Energy and environmental economics; machine learning in economics; digital economy; agricultural economics; international trade; economic complexity.

PUBLICATIONS

Bold = author; * = corresponding author. Indexing, JCR quartile and impact factor (IF) from Clarivate Journal Citation Reports 2026 release (2025 data).

Peer-Reviewed Journal Articles

1. **Qayyum, M.**, Bashir, M. A., Ali, M., & Mehmood, U. (2026). A path to carbon neutrality: Exploring the impact of natural resource use on CO2 emissions and load capacity factor in BICS. *Energy & Environment*. Advance online publication. https://doi.org/10.1177/0958305X261474724 [SSCI, Q1; IF 5.1]
2. **Qayyum, M.**, Ali, M., Hassan, M. W., et al. (2026). Harnessing machine learning to decode the energy–economy–environment trilemma: Insights from renewable and non-renewable energy use in Pakistan. *Environment, Development and Sustainability*. Advance online publication. https://doi.org/10.1007/s10668-026-07985-z [SCIE, Q2; IF 4.9]
3. Yu, Y., **Qayyum, M.***, & Li, S. (2024). Trade dynamics of environmental goods within global energy economy and their impacts on green technological innovation: A complex network analysis. *Energy Economics*, 107957. https://doi.org/10.1016/j.eneco.2024.107957 [SSCI, Q1; IF 13.5]

4. **Qayyum, M.**, Zhang, Y., Ali, M., & Kirikkaleli, D. (2024). Towards environmental sustainability: The role of information and communication technology and institutional quality on ecological footprint in MERCOSUR nations. *Environmental Technology & Innovation*, 103523. https://doi.org/10.1016/j.eti.2023.103523 [SCIE, Q1; IF 7.3]
5. Zhang, Y., **Qayyum, M.***, & Yu, Y. (2023). Effects of resource abundance on economic complexity: Evidence from spatial panel model. *Journal of Cleaner Production*, 139134. https://doi.org/10.1016/j.jclepro.2023.139134 [SCIE, Q1; IF 10.7]
6. **Qayyum, M.**, Zhang, Y., Wang, M., Yu, Y., Li, S., Ahmad, W., Maodaa, S. N., Sayed, S. R. M., & Gan, J. (2023). Advancements in technology and innovation for sustainable agriculture: Understanding and mitigating greenhouse gas emissions from agricultural soils. *Journal of Environmental Management*, *347*, 119147. https://doi.org/10.1016/j.jenvman.2023.119147 [SCIE, Q1; IF 9.2]
7. **Qayyum, M.**, Li, S., Jian, G., & Yu, Y. (2023). Evaluating the impacts of imported cultural goods on carbon emission: Cross-country evidence. *Energy & Environment*. https://doi.org/10.1177/0958305X231192358 [SSCI, Q1; IF 5.1]
8. **Qayyum, M.***, Yu, Y., Bhatti, U. A., & Li, S. (2023). Evaluation of the One Belt and One Road (OBOR) in economic development and suggestions analysis based on SWOT analysis with weighted AHP and entropy methods. *Multimedia Tools and Applications*, *82*, 14985–15006. https://doi.org/10.1007/s11042-022-13565-w
9. **Qayyum, M.**, Yu, Y., Tu, T., Nizamani, M. M., Ahmad, A., & Ali, M. (2022). Relationship between economic liberalization and intellectual property protection with regional innovation in China: A case study of Chinese provinces. *PLoS ONE*, *17*(1), e0259170. https://doi.org/10.1371/journal.pone.0259170 [SCIE, Q2; IF 2.8]
10. **Qayyum, M.**, Yu, Y., Nizamani, M. M., Raza, S., Ali, M., & Li, S. (2022). Financial instability and CO2 emissions in India: Evidence from ARDL bound testing approach. *Energy & Environment*. https://doi.org/10.1177/0958305X211065019 [SSCI, Q1; IF 5.1]
11. Yu, Y., & **Qayyum, M.*** (2022). Dynamics between carbon emission, imported cultural goods, human capital, income and energy consumption: Renewed evidence from panel VAR approach. *Environmental Science and Pollution Research*. https://doi.org/10.1007/s11356-022-19862-z
12. **Qayyum, M.**, Ali, M., Nizamani, M. M., Li, S., Yu, Y., & Jahanger, A. (2021). Nexus between financial development, renewable energy consumption, technological innovations and CO2 emissions: The case of India. *Energies*, *14*(15), 4505. https://doi.org/10.3390/en14154505 [SCIE, Q3; IF 3.9]
13. **Qayyum, M.**, Yu, Y., & Li, S. (2021). The impact of economic complexity on embodied carbon emission in trade: New empirical evidence from cross-country panel data. *Environmental Science and Pollution Research*. https://doi.org/10.1007/s11356-021-14414-3
14. Yu, Y., & **Qayyum, M.*** (2021). Impacts of financial openness on economic complexity: Cross-country evidence. *International Journal of Finance & Economics*. https://doi.org/10.1002/ijfe.2491 [SSCI, Q1; IF 3.7]
15. Sun, P., Liu, L., & **Qayyum, M.** (2021). Energy efficiency comparison amongst service industry in Chinese provinces from the perspective of heterogeneous resource endowment: Analysis using undesirable super efficiency SBM-ML model. *Journal of Cleaner Production*, *328*, 129535. https://doi.org/10.1016/j.jclepro.2021.129535 [SCIE, Q1; IF 10.7]
16. Yang, B., Ali, M., Nazir, M. R., Ullah, W., & **Qayyum, M.** (2020). Financial instability and CO2 emissions: Cross-country evidence. *Air Quality, Atmosphere & Health*, *13*(4), 459–468. https://doi.org/10.1007/s11869-020-00809-7 [SCIE, Q3; IF 3.1]

Conference Proceedings

17. Younas, N., Asghar, Z., **Qayyum, M.**, & Khan, F. (2017). Education and socio-economic factors impact on earning for Pakistan—A big data analysis. In *Future Intelligent Vehicular Technologies: First International Conference, Future 5V 2016, Porto, Portugal, September 15, 2016, Revised Selected Papers* (Vol. 185, p. 215). Springer.

Book Chapters

18. Nizamani, M. M., Awais, M., **Qayyum, M.**, Tarafder, E., Khaskheli, M. A., & Wang, Y. (2025). Carbon emission pattern analysis and its relationship with climate change. In *Deep Learning for Earth Observation and Climate Monitoring* (pp. 63–94). Elsevier. https://doi.org/10.1016/B978-0-443-24712-5.00005-1
19. Nizamani, M. M., Awais, M., **Qayyum, M.**, Khaskheli, M. A., Tarafder, E., & Wang, Y. (2025). The application of geographic information systems and remote sensing technologies in urban ecology. In *Deep Learning for Earth Observation and Climate Monitoring* (pp. 137–163). Elsevier. https://doi.org/10.1016/B978-0-443-24712-5.00008-7
20. Nizamani, M. M., Muhae-Ud-Din, G., Zhang, Q., Awais, M., **Qayyum, M.**, Farhan, M., ... & Wang, Y. (2024). Application of machine learning to urban ecology. In *Deep Learning for Multimedia Processing Applications* (pp. 209–245). CRC Press. https://doi.org/10.1201/9781032646268-10
21. Nizamani, M. M., Zhang, Q., Muhae-Ud-Din, G., Awais, M., **Qayyum, M.**, Farhan, M., ... & Wang, Y. (2024). Application of GIS and remote-sensing technology in ecosystem services and biodiversity conservation. In *Deep Learning for Multimedia Processing Applications* (pp. 284–321). CRC Press. https://doi.org/10.1201/9781032646268-12

GRANTS AND RESEARCH PROJECTS

Principal Investigator 2026 – 2028

Fujian Provincial Department of Education, Education and Research Project for Young and Middle-aged Teachers (Science & Technology), No. JAT251221: “Strategic Choices for Industrial Upgrading in Fujian Province: Opportunities in the China–Pakistan Economic Corridor Based on Product Space.” RMB 100,000. Ongoing.

Principal Investigator May 2024 – Dec 2024

Guangzhou Municipal Finance, Postdoctoral Research Project Funding, No. 624021-78: “The Role of Artificial Intelligence Development and Economic Growth for Sustainable Agriculture in Asian Countries: Do Carbon Emissions and Deforestation Affect Agricultural Productivity?” RMB 200,000. Completed.

Participant 2022 – 2025

National Natural Science Foundation of China (NSFC), Regional Science Fund Project, No. 72164008: “Research on Total Factor Energy Efficiency Improvement Driven by Ecological Constraints and Foreign Investment: An Analytical Framework Based on the Energy–Economy–Environment System.” RMB 280,000. Completed.

TEACHING EXPERIENCE

Quanzhou University of Information Engineering 2025 – present

Associate Professor, School of Economics and Management
Courses taught: International Trade Theory and Practice; Management Studies; Macroeconomics

Hubei University of Education, Wuhan, China Sep 2020 – Jun 2021; Sep – Dec 2022

Lecturer in Economics (volunteer, online)
Economics, International Trade, and Business English (12 teaching hours per week in 2020–2021); prepared lectures, exams and assignments

HONORS, AWARDS AND FELLOWSHIPS

Chinese Government Scholarship for doctoral study, China Scholarship Council 2016 – 2020

Award, International Graduate Student Innovative Research Project, Zhongnan University of Economics and Law 2017

Certificate of Commendation for Active Participation in Chinese Language Learning, Zhongnan University of Economics and Law 2017

HEC Scholarship for M.Phil. study, Higher Education Commission of Pakistan 2013 – 2016

Outstanding Young Talent, Government of Pakistan 2012

District Merit Scholarship for B.Sc. study, University of Agriculture, Faisalabad 2009 – 2013

SERVICE

Peer reviewer for more than 15 high-level international journals

PROFESSIONAL MEMBERSHIPS

Member, Young Economists Association, Pakistan Institute of Development Economics (2014)

SKILLS

Econometrics and programming: R, RStudio, R Markdown; Stata; Python (basic); IBM SPSS; Origin (advanced)

GIS and visualization: ArcGIS Desktop/Pro (advanced); Adobe Photoshop, Illustrator

Languages: English (proficient; IELTS 7.0); Urdu and Punjabi (native). Chinese: fluent spoken (occasionally teaches in Chinese); reading intermediate; writing basic

REFERENCES

Prof. Zhang Yanping
Postdoctoral Supervisor; Dean of Management Department, Guangzhou University, Guangzhou, China

Uzair Aslam Bhatti
Associate Professor, Hainan University, Haikou, China

Contact details available upon request.